

Enterprise Account Decision-Chain Mapping & Cross-Team Intelligence Aggregation Assistant

Final Project Analysis

Wen Hao | PE6201 Emerging AI Technologies | 1 October 2026 | Word count: 1200

1 Problem and significance

Enterprise account teams often hold customer intelligence in separate CRM notes, email threads, meeting records, calendars and support tickets. A Key Account Manager preparing for a client meeting may spend valuable time reconstructing who influences the decision, who can approve the purchase, what changed recently and which risk requires attention. Fragmentation can produce a briefing that is slow, incomplete or overconfident. AccountLens addresses this bounded preparation task: one account enters the workflow and one evidence-grounded Account Panorama Briefing comes out for human review.

AccountLens does not remove the manager from the decision. It shifts work from assembling records to verifying one evidence-linked hypothesis set. In the blinded pilot, this shift reduced median preparation time by 40 seconds, or 21.6%, but fictional accounts and self-reported timing prevent a direct productivity or revenue claim.

2 Solution and scope

The prototype infers six decision-role classes, highlights a likely commercial signatory, maps evidence-linked workflow hypotheses, ranks activity from the previous 30 days, identifies unresolved tickets, unanswered email and competitor signals, and recommends no more than two next actions. It provides a local dashboard plus one-page PDF and Slack-ready Markdown exports. It does not connect to live Salesforce, Outlook, Jira or Slack, and it never sends a message or changes a source system. This keeps the project focused on the teacher's requested unit of value: one account and one meeting briefing.

Under the Class 4 definition, AccountLens is a bounded AI-assisted workflow rather than a full agent. Application code fixes the sequence and performs filtering, deterministic risk checks, validation and export. The model performs one structured inference; it does not choose tools, vary the action sequence or receive observations in a thought-action loop. A full agent would add cost, compounded failure risk and governance exposure without improving the measured task.

3 Data and implementation

Because real enterprise systems were unavailable, the project uses reproducible fictional data. A fixed-seed generator creates 60 accounts for development, validation and frozen testing. A separate teacher-aligned set contains 10 hand-authored accounts. Each includes two email threads with explicit CC lists, one meeting attendee list, two support tickets, one CRM note, six contacts and a `true_signatory_contact_id`. The signatory is defined as the contact with final commercial approval authority. This label is removed before every model call and used only for offline scoring.

The application rents a GPT-4o-compatible model and provider infrastructure while owning the data schema, prompts, orchestration, evidence validation, evaluation and governance rules. Exact account filtering is sufficient, so a vector database and RAG layer were not added. Structured output is checked against a Pydantic schema; invalid contact or evidence IDs trigger retry or visible failure. Twenty-four automated tests cover data generation, leakage prevention, role metrics, evidence validation, reporting and economics.

4 Evaluation method

The primary comparison is a deterministic rules baseline, B0, against the evidence-grounded model, M2. The 60-account dataset was split by account into 30 development, 10 validation and 20 test accounts. Model name, prompt and confidence threshold were frozen before one final test execution. Metrics include Macro-F1, accuracy, evidence precision, critical-event recall, abstention, p95 latency and estimated API cost. The supplementary 10-account evaluation measures signatory precision, recall and selection rate so that omissions cannot be hidden behind precision alone.

Five anonymous participants also completed a counterbalanced blinded comparison of B0 and M2 briefings for four fictional validation accounts. They recorded preparation time, six five-point ratings and a preference. The

study is reported descriptively because the sample is small, responses were self-reported and participant identity or live sessions were not independently verified.

5 Results

On the frozen 20-account test, M2 achieved 1.000 accuracy and Macro-F1, compared with B0 Macro-F1 of 0.7024. M2 evidence precision and critical-event recall were both 1.000, with p95 latency of 8.13 seconds. On the separate teacher-aligned set, B0 selected seven signatories and all seven were correct, giving 1.00 precision but only 0.70 recall and selection rate. M2 selected all 10 and matched all 10 labels, giving 1.00 precision, recall and selection rate. These are controlled fictional-data results, not proof of real-world accuracy.

The frozen model run cost USD 0.321290 for 20 briefings, or USD 0.016065 per briefing. The teacher-aligned run cost USD 0.125933. In the blinded pilot, M2 was preferred in 20 of 20 paired comparisons, median preparation time fell from 185 to 145 seconds, and the median mean rating increased from 2.83 to 4.50. The uniform small sample makes the result directional rather than population-level evidence.

6 Tuning, business and technical trade-offs

Prompt v1 misclassified three champions as unknown by conflating influence with final purchase authority. I added behavioural role definitions and separated advocacy from budget approval; development Macro-F1 rose from 0.8901 to 1.000. M2 then matched H1's validation quality with lower estimated cost and p95 latency, so I rejected the more complex hybrid. Thresholds from 0.50 to 0.85 tied; retaining 0.65 avoided tuning to a meaningless validation difference.

The likely signatory is a Class 5 Archetype B output because the system manufactures a measurement that source systems do not directly record; briefing summarisation is an Archetype A subtask. The use case passes the prototype gates for worth, possibility, affordability, absorption and killability, but production affordability is unproven. The observed variable cost excludes integration, monitoring, maintenance, human review and governance. At the variable-only rate, a USD 10 balance supports about 622 briefings, but this is not a production budget.

A future pilot should stop or be redesigned if signatory precision is below 0.85, selection rate below 0.70, evidence precision below 0.90, unsupported-claim rate above 0.05, p95 latency above 15 seconds, or full cost exceeds the measured value of preparation time saved.

7 Responsible use and limitations

The system is advisory. Confidence, abstention, evidence IDs, missing-role flags and human confirmation reduce the risk of redirecting account effort based on an unsupported role. Workflow edges are labelled as hypotheses rather than verified reporting lines. Source records are treated as untrusted data, raw content is hidden by default, logs contain metadata only, and no outbound write is available.

The main limitation is synthetic data. The accounts are balanced and generated from repeated patterns, so the perfect M2 score does not establish performance across real organisations, languages, writing styles or data-quality failures. Production use would require lawful access, source provenance, independent signatory annotation, named data ownership, role-based access, retention controls, drift monitoring and a larger observed user study linked to downstream decision quality.

8 Conclusion

AccountLens demonstrates that a narrow, evidence-grounded workflow can turn fragmented account records into a faster and more actionable meeting briefing while preserving human authority. The contribution is not autonomy; it is a measurable combination of signatory selection, role coverage, evidence traceability, abstention, bounded recommendations and explicit failure handling. The prototype meets the course objective on fictional data and defines clear conditions that must be satisfied before any real-data pilot or production claim. The next step is a privacy-approved, read-only pilot with de-identified records, independent labels and observed user sessions; autonomy should wait until those gates are met.